

Abrikosov Lattices in the Unit Disk

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Abstract

We address part (v) of Vicențiu D. Rădulescu’s 2006 *SIAM Review* Proposed Problem 06-006, “Abrikosov Lattices in Superconductivity” [Răd06]. The problem introduces a logarithmic renormalized energy for point configurations in the open unit disk and treats the cases of two and three points. Its remaining question is to identify the minimizers for $n \geq 4$ and determine whether they approach the boundary as $n \rightarrow \infty$.

After reversing the sign and passing to an equivalent multiplicative objective, the problem separates into two sharp estimates: a Cauchy-kernel determinant bound and a weighted Vandermonde bound. The second reduces the radial part to a one-variable inequality. Its equality case fixes all radii, while equality in Hadamard’s inequality fixes the angular spacing. The maximizers are exactly the centered regular n -gons, up to rotation and relabeling, with radius

$$\rho_n = \left(\frac{n-1}{3n-1} \right)^{1/(2n)}.$$

We also obtain the sharp value of the multiplicative objective and show that the minimum modulus a_n of every maximizing configuration satisfies $a_n = \rho_n \rightarrow 1$ as $n \rightarrow \infty$. The finite and algebraic steps of the argument have also been formalized in Lean 4 with Mathlib [dMU21, The20].

1 Introduction and the original problem

Rădulescu’s Proposed Problem 06-006 asks in part (v) for the minimizing configuration when $n \geq 4$ and whether such configurations approach the boundary [Răd06]. Precisely, if $z_1^{(n)}, \dots, z_n^{(n)}$ is a minimizing configuration and

$$a_n = \min_{1 \leq j \leq n} |z_j^{(n)}|,$$

does $a_n \rightarrow 1$?

Based on the literature search conducted for this project, I did not locate a published resolution of this exact part. It sits within a broader family of extremal problems involving Vandermonde products and weighted Fekete configurations, including fixed-set problems on the unit circle [DP20]. Related product inequalities go back to Cassels and Alexander [Cas66, Ale72], and regular-polygon extremality appears in more recent work [Dub20, Oun26]. These results concern related products, or settings in which the radius is fixed or all points lie on the unit circle. They do not by themselves determine the extremizer when both radii and angles are free. The argument below is directed at the specific question in part (v); it makes no priority claim beyond that literature search.

I use the negative of Rădulescu’s energy, divided by π . This turns his minimization problem into the maximization problem for W below without changing either the extremizing configurations or the boundary question. The theorem in fact determines the radius and all extremizers for every $n \geq 2$, including the cases already treated in the proposed problem.

2 Main theorem and proof

Let $n \geq 2$ and $z_1, \dots, z_n$ lie in the open unit disk $\mathbb{D}$. Consider

$$W(z_1, \dots, z_n) = 2 \sum_{i < j} \log |z_i - z_j| + \sum_{i,j=1}^n \log |1 - z_i \bar{z}_j|. \quad (1)$$

We set $W = -\infty$ if two points coincide. Define

$$q_n = \frac{n-1}{3n-1}, \quad \rho_n = q_n^{1/(2n)}, \quad M_n = n^n q_n^{(n-1)/2} (1 - q_n)^n. \quad (2)$$

Theorem 2.1. *For $z_1, \dots, z_n \in \mathbb{D}$, one has $\exp W(z_1, \dots, z_n) \leq M_n$. Equality holds precisely when, in some order,*

$$z_j = \rho_n e^{i(\theta + 2\pi j/n)}, \quad 0 \leq j < n,$$

for some $\theta \in \mathbb{R}$. Consequently, if $a_n = \min_j |z_j|$ for any choice of a maximizing n -tuple, then $a_n = \rho_n \rightarrow 1$.

The proof uses the classical Cauchy and Vandermonde determinant formulas, unitary Schur decomposition, and Hadamard's inequality in their standard forms. For the matrix-analytic facts, see [HJ13]. The problem-specific estimates and equality analysis are given below.

2.1 Multiplicative reformulation

Let

$$V = (z_i^k)_{\substack{1 \leq i \leq n, \\ 0 \leq k < n}}, \quad \Delta = \det V = \prod_{i < j} (z_j - z_i),$$

and put

$$B(z) = \prod_{i,j} |1 - z_i \bar{z}_j|, \quad R(z) = \prod_i (1 - |z_i|^{2n}), \quad F(z) = |\Delta|^2 B(z).$$

Exponentiating (1) gives exactly F . Thus maximizing W and F are equivalent; coincident points have $F = 0$ and cannot maximize.

2.2 The first determinant estimate

Lemma 2.2. *For every $z_1, \dots, z_n \in \mathbb{D}$,*

$$B(z) \leq R(z). \quad (3)$$

Proof. First suppose the points are distinct. Introduce

$$K = \left(\frac{1}{1 - z_i \bar{z}_j} \right)_{i,j}, \quad G = VV^* = \left(\sum_{k=0}^{n-1} (z_i \bar{z}_j)^k \right)_{i,j}, \quad Z = \text{diag}(z_1^n, \dots, z_n^n).$$

The geometric-series identity gives

$$G = K - ZKZ^*. \quad (4)$$

The Cauchy determinant formula, specialized to this kernel, gives

$$|\det K| B(z) = |\Delta|^2 = |\det G|. \quad (5)$$

We need K to be positive definite, not merely positive semidefinite. For $c = (c_1, \dots, c_n) \in \mathbb{C}^n$, expansion of the geometric series gives

$$c^* K c = \sum_{m \geq 0} \left| \sum_{i=1}^n \bar{c}_i z_i^m \right|^2. \quad (6)$$

The interchange is harmless because $|z_i| < 1$ and the sum is finite in i . If (6) is zero, then every summand is zero. In particular, the first n moments, for $0 \leq m < n$, vanish. These equations form the Vandermonde system $V^* c = 0$ (up to the harmless choice of conjugation convention). Since the z_i are distinct, $\det V = \Delta \neq 0$, and hence $c = 0$. Thus K is positive definite; in particular, its positive square root and $K^{-1/2}$ are well-defined.

Set $S = K^{-1/2}$ and $T = SZK^{1/2}$. Multiplying (4) by S on both sides gives

$$I - TT^* = SGS \succeq 0. \quad (7)$$

Taking determinants makes the normalization explicit. Since K is positive definite, $\det K > 0$ and

$$\begin{aligned} |\det(SGS)| &= |\det S|^2 |\det G| \\ &= \frac{|\det G|}{\det K} \\ &= B(z), \end{aligned}$$

where the second equality uses $S = K^{-1/2}$ and the third uses (5). Together with (7), this yields

$$B(z) = |\det(I - TT^*)|. \quad (8)$$

If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of T , then

$$|\det(I - TT^*)| \leq \prod_i (1 - |\lambda_i|^2). \quad (9)$$

Indeed, take a unitary Schur decomposition $T = URU^*$ with R upper triangular. Unitary conjugation of (7) gives $I - RR^* \succeq 0$. Hadamard's inequality, applied to this positive semidefinite matrix, gives

$$\begin{aligned} \det(I - RR^*) &\leq \prod_i \left(1 - \sum_j |R_{ij}|^2 \right) \\ &\leq \prod_i (1 - |R_{ii}|^2). \end{aligned}$$

The individual factors on both lines are nonnegative: this follows from the nonnegativity of the diagonal entries and, on the second line, from the presence of the $j = i$ summand. Since $R_{ii} = \lambda_i$ and determinant is unchanged by unitary conjugation, this proves (9). Since T is similar to Z , its eigenvalues are $z_1^n, \dots, z_n^n$. Equations (8) and (9) prove (3) for distinct points. Distinct tuples are dense in $\mathbb{D}^n$, and both sides are continuous, so the general case follows. $\square$

2.3 The scalar inequality

The definition (2) gives $\rho_n^{2n} = q_n$. The radial part of the argument is contained in the following sharp one-variable estimate.

Lemma 2.3. *For $t \geq 0$, set*

$$h_n(t) = (1 - q_n t^n)(1 + t + \cdots + t^{n-1}).$$

Then $h_n(t) \leq n(1 - q_n)$, with equality if and only if $t = 1$.

Proof. Write $S(t) = 1 + t + \cdots + t^{n-1}$. Because $q_n = (n-1)/(3n-1)$, direct differentiation first gives

$$(3n-1)h'_n(t) = (3n-1)S'(t) - (n-1)(t^n S'(t) + nt^{n-1}S(t)). \quad (10)$$

Collecting powers of t in (10), and then factoring out $1-t$, gives

$$(3n-1)h'_n(t) = (1-t)Q_n(t), \quad Q_n(t) = \sum_{m=0}^{2n-3} b_m t^m, \quad (11)$$

where

$$b_m = \begin{cases} \frac{3n-1}{2}(m+1)(m+2), & 0 \leq m \leq n-2, \\ \frac{n-1}{2}(2n(2n-1) - (m+1)(m+2)), & n-1 \leq m \leq 2n-3. \end{cases}$$

Every b_m is strictly positive. This is immediate in the first range. In the second range, $m+1 \leq 2n-2 < 2n$ and $m+2 \leq 2n-1$, so $(m+1)(m+2) < 2n(2n-1)$. Hence $Q_n(t) > 0$ for $t \geq 0$. Formula (11) shows that h_n is strictly increasing on $[0, 1]$ and strictly decreasing on $[1, \infty)$. Its unique maximum is therefore $h_n(1) = n(1 - q_n)$, proving both assertions. $\square$

2.4 The weighted Vandermonde bound and equality

Lemma 2.4. *For $z_1, \dots, z_n \in \mathbb{D}$,*

$$|\Delta|^2 R(z) \leq n^n q_n^{(n-1)/2} (1 - q_n)^n = M_n. \quad (12)$$

Equality forces the points to be a centered regular n -gon of radius ρ_n .

Proof. Consider the weighted Vandermonde matrix

$$A_{ik} = \sqrt{1 - |z_i|^{2n}} \left(\frac{z_i}{\rho_n} \right)^k, \quad 1 \leq i \leq n, \quad 0 \leq k < n. \quad (13)$$

Factoring from its rows and columns gives

$$|\det A|^2 = \rho_n^{-n(n-1)} |\Delta|^2 R(z). \quad (14)$$

Put $t_i = |z_i|^2 / \rho_n^2$. The squared norm of row i is

$$\begin{aligned} \sum_{k=0}^{n-1} |A_{ik}|^2 &= (1 - |z_i|^{2n}) \sum_{k=0}^{n-1} \left(\frac{|z_i|^2}{\rho_n^2} \right)^k \\ &= (1 - q_n t_i^n) \sum_{k=0}^{n-1} t_i^k = h_n(t_i) \leq n(1 - q_n) \end{aligned}$$

by Lemma 2.3. Hadamard's inequality and (14) imply

$$\rho_n^{-n(n-1)} |\Delta|^2 R(z) \leq \prod_i \sum_k |A_{ik}|^2 \leq (n(1 - q_n))^n.$$

Since $\rho_n^{n(n-1)} = q_n^{(n-1)/2}$, this is (12).

It remains to identify the equality case. Suppose equality holds in (12). In the chain

$$|\det A|^2 \leq \prod_i \|A_{i\bullet}\|_2^2 \leq (n(1 - q_n))^n,$$

both inequalities must then be equalities. Each row norm squared is nonnegative and bounded above by the positive number $n(1 - q_n)$. If even one factor were strictly smaller, their product would be strictly smaller as well. Thus every row-factor bound is an equality:

$$h_n(t_i) = n(1 - q_n) \quad (1 \leq i \leq n).$$

The equality case of Lemma 2.3 gives $t_i = 1$, hence

$$|z_i| = \rho_n \quad (1 \leq i \leq n). \quad (15)$$

In particular, every row has squared norm $n(1 - q_n) > 0$.

Equality in the first inequality is the equality case of Hadamard's determinant inequality. Thus the nonzero rows of A are pairwise orthogonal. For $i \neq j$, their inner product is

$$(1 - \rho_n^{2n}) \sum_{k=0}^{n-1} \left(\frac{z_i \bar{z}_j}{\rho_n^2} \right)^k.$$

Since $0 < \rho_n < 1$, the prefactor is nonzero, and orthogonality therefore implies

$$\sum_{k=0}^{n-1} \left(\frac{z_i \bar{z}_j}{\rho_n^2} \right)^k = 0. \quad (16)$$

By (15), the number in parentheses is z_i/z_j . Writing $\xi = z_i/z_j$, the geometric-series identity $(1 - \xi) \sum_{k=0}^{n-1} \xi^k = 1 - \xi^n$ applied to (16) gives $\xi^n = 1$. Moreover $\xi \neq 1$, since otherwise the sum in (16) would equal n . Thus every ratio of two distinctly indexed points is a nontrivial n th root of unity.

Finally, equality gives $|\det A|^2 = (n(1 - q_n))^n > 0$, so A is invertible. Two equal z_i would give two equal rows in (13), contradicting invertibility; hence the points are distinct. Fix z_1 . The n quotients z_i/z_1 are therefore distinct n th roots of unity. There are exactly n such roots, so these quotients exhaust all of them. Hence, after a permutation,

$$z_j = z_1 e^{2\pi i j/n} = \rho_n e^{i(\theta + 2\pi j/n)}.$$

This proves the equality classification. □

2.5 Sharpness and the limiting radius

By Lemmas 2.2 and 2.4,

$$F(z) = |\Delta|^2 B(z) \leq |\Delta|^2 R(z) \leq M_n.$$

The equality analysis above shows that only the regular configuration can maximize F . Conversely, let $\omega = e^{2\pi i/n}$ and $z_j = \rho_n e^{i\theta} \omega^j$. With $\mathcal{F} = (\omega^{jk})_{0 \leq j, k < n}$ and $D = \text{diag}((\rho_n e^{i\theta})^k)_{0 \leq k < n}$, the Vandermonde matrix is $V = \mathcal{F}D$. The standard Fourier identity $\mathcal{F}\mathcal{F}^* = nI$ gives $|\det \mathcal{F}|^2 = n^n$, while

$$|\det D|^2 = \prod_{k=0}^{n-1} \rho_n^{2k} = \rho_n^{n(n-1)}.$$

Consequently,

$$|\Delta|^2 = n^n \rho_n^{n(n-1)}. \quad (17)$$

For fixed i , the numbers ω^{i-j} run through all n th roots of unity. The identity $\prod_{j=0}^{n-1} (x - \omega^j) = x^n - 1$ gives

$$\prod_{j=0}^{n-1} (1 - z_i \bar{z}_j) = \prod_{j=0}^{n-1} (1 - \rho_n^2 \omega^{i-j}) = 1 - \rho_n^{2n} > 0.$$

Multiplication over i yields

$$B(z) = (1 - \rho_n^{2n})^n = (1 - q_n)^n. \quad (18)$$

Together, (17) and (18) give $F(z) = M_n$. This proves sharpness and completes Theorem 2.1.

Finally, every maximizer has $|z_j| = \rho_n$, so $a_n = \rho_n$. Since $q_n \rightarrow 1/3$,

$$\log \rho_n = \frac{\log q_n}{2n} \rightarrow 0.$$

Exponentiating gives $a_n = \rho_n \rightarrow 1$.

3 Formal verification

The argument has been formalized in Lean 4 with Mathlib [dMU21, The20]. The main file

`RomanianOpenProblemStrict.lean`

checked successfully with Lean exit code 0. Its principal declarations are

- `objective_le_upperBound` for the sharp upper bound;
- `objective_eq_upperBound_iff_regular` for the full equality classification; and
- `rho_tendsto_one` for the limiting radius.

For each declaration, `#print axioms` reports only `propext`, `Classical.choice`, and `Quot.sound`, with no `sorryAx` or custom axioms. The checked Lean project is available as supplementary material in the accompanying repository/archive. It is not embedded in the manuscript, which instead gives the conventional human-readable proof.

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